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Development of an in-the-wild hand gesture recognition model using synthetic multi-view images

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Major: Digital Communication and Multimedia Engineering

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1. Introduction

1.1 Motivation and Problem statement

Canonical / Controlled View

- Frontal camera
- Canonical view
- Clean landmarks
- Guided pose

Hand-in-the-Wild / Natural View

- Non-canonical viewpoint
- Arbitrary hand orientation
- Self-occlusion
- Noisy landmarks

High accuracy under canonical views **does not guarantee robustness** under natural arbitrary viewpoints.

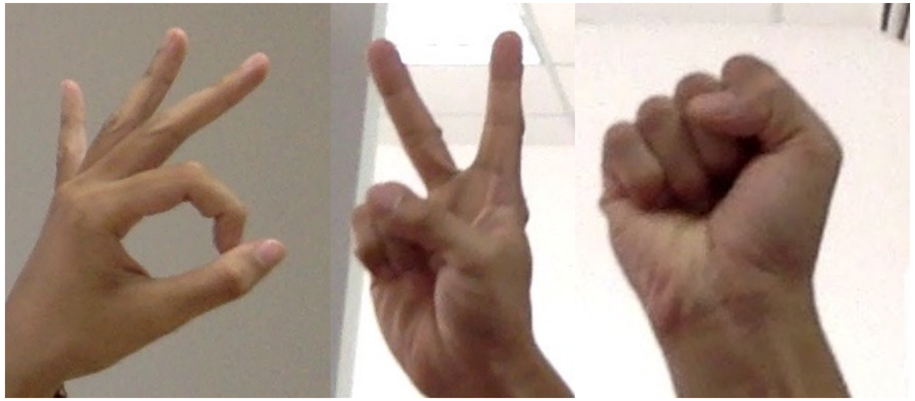
1. Introduction

1.1 Motivation and Problem statement

Core challenge: Hand gestures are highly sensitive to viewpoint changes.

Same gesture, different viewpoints:

- The same gesture may produce different skeleton distributions.
- Finger visibility changes across camera angles.
- Hand orientation affects the observed landmark structure.



Thumbdown

May resemble Rock under certain viewpoints

Thumbup

May resemble Rock under occlusion

Scissors

May resemble Thumb gesture under viewpoint variation

Research question:

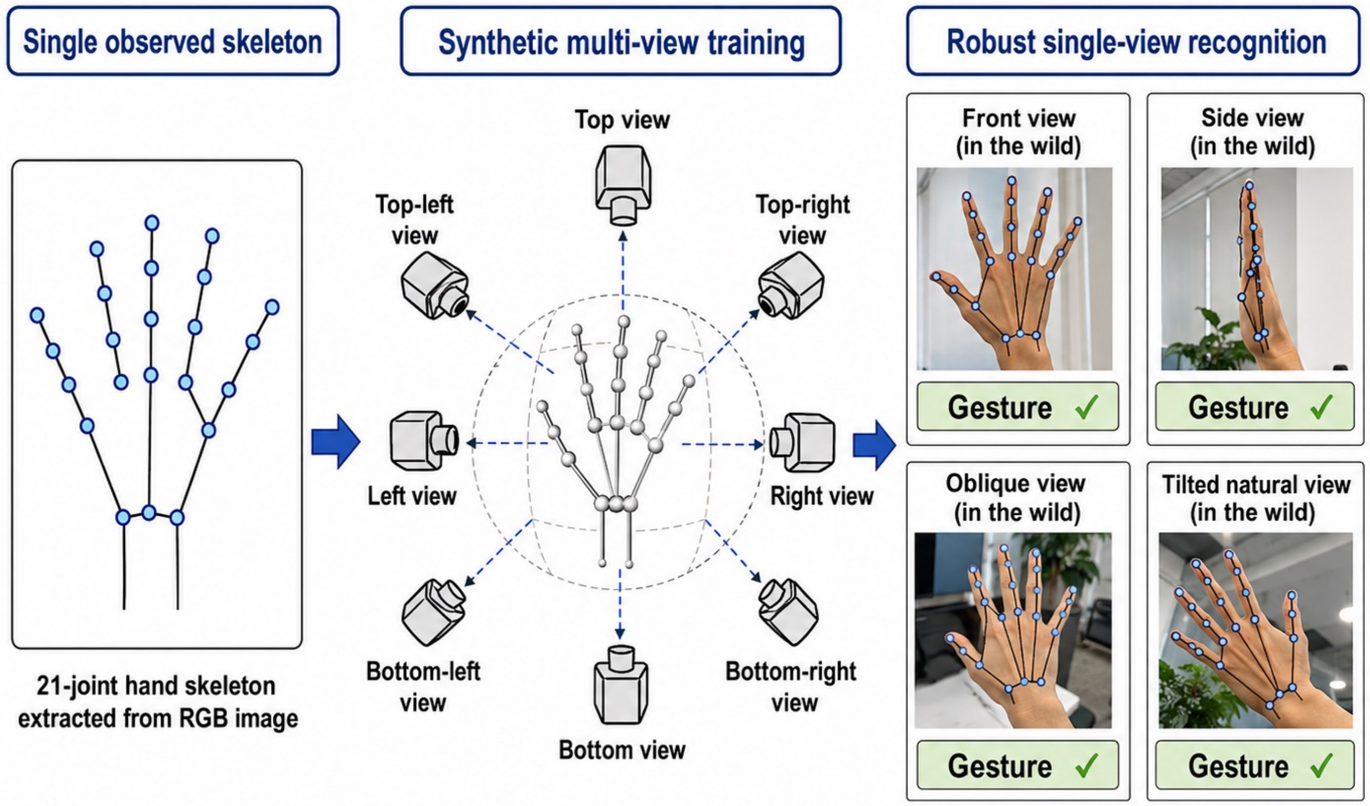
Can synthetic multi-view observations improve viewpoint robustness while preserving single-view inference?

1.2 Research objectives

Objective: Develop a hand gesture recognition model that is robust to natural viewpoint variations.

Key Ideas:

- Reconstructs a plausible 3D hand pose,
- Generates multiple virtual viewpoints in Blender,
- And uses these observations to improve gesture representation learning



Expected Output: A single-view gesture recognition model that can maintain good performance across different natural hand viewpoints.

2. Related Work

Existing direction	Key limitation
RGB-based recognition [1]	Sensitive to background, lighting, and appearance domain gap
Skeleton-based recognition [2]	Compact, but still viewpoint-sensitive
2D augmentation / synthetic data [3]	Increases data diversity, but may not model true 3D viewpoint changes
Real multi-view learning [4]	Requires synchronized cameras or multi-view input

Research gap:

Improve viewpoint robustness without real multi-camera data and preserve single-skeleton inference.

This thesis:

Synthetic camera-space skeleton views during training

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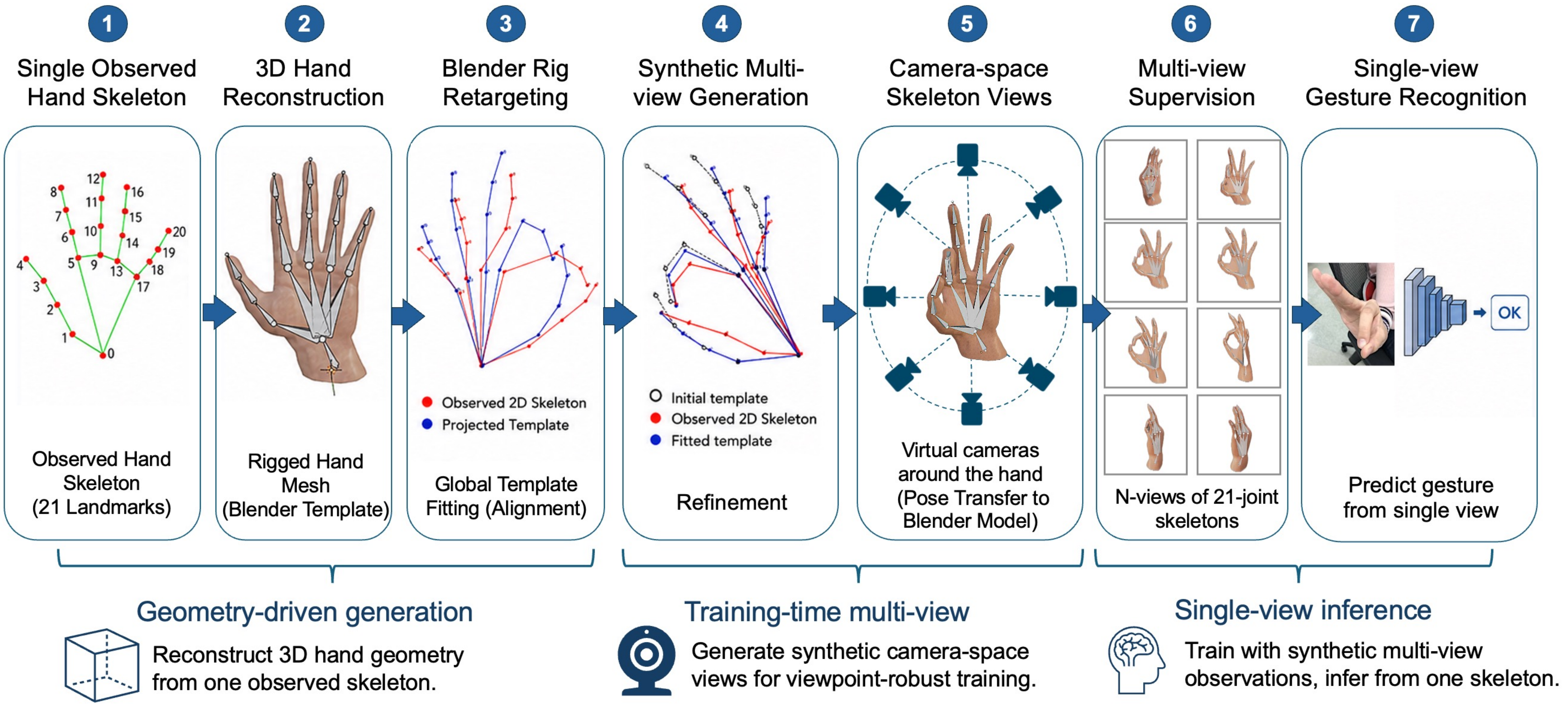
3. Proposed method

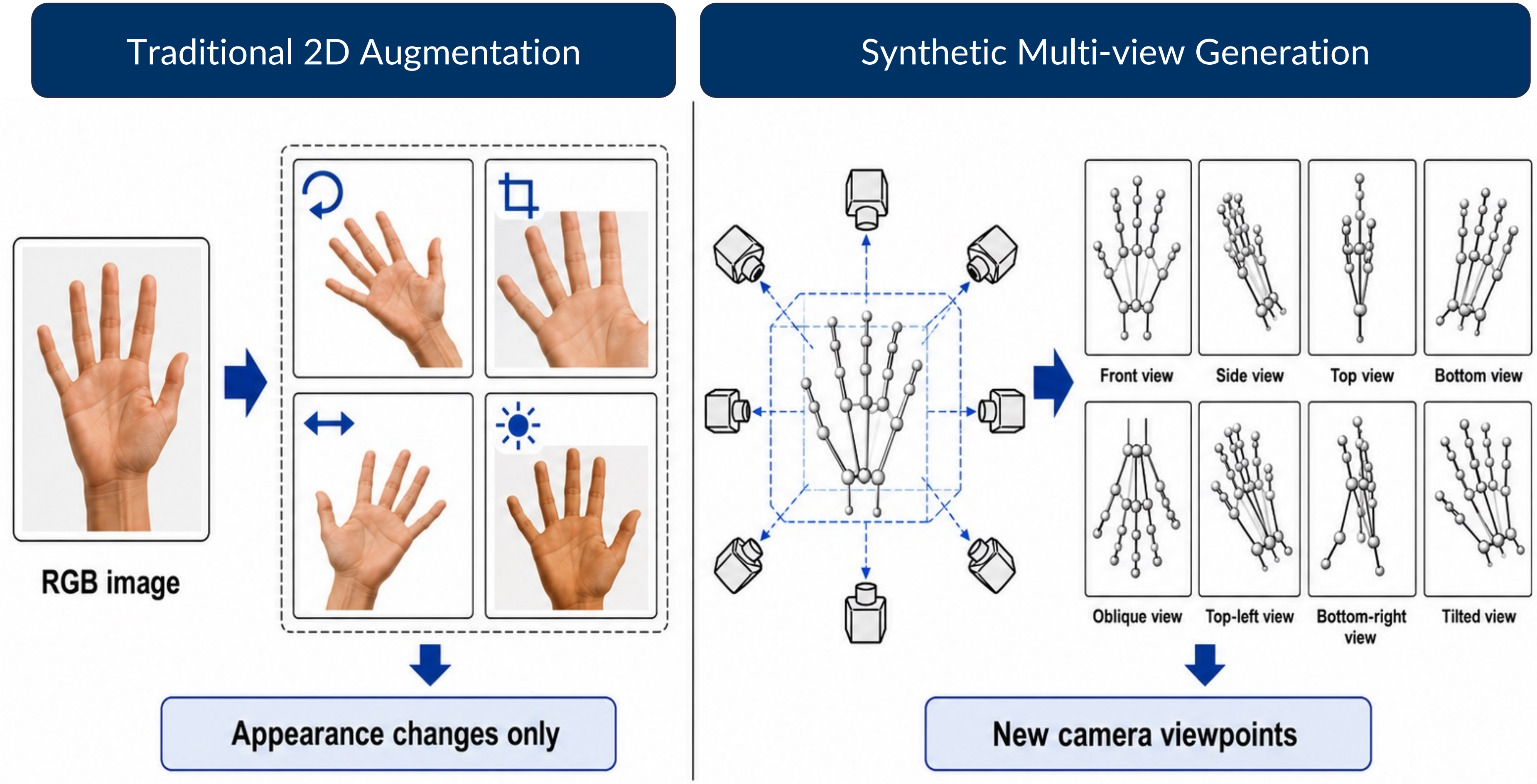
3.1 Overall framework



3.2 Why synthetic multi-view?

Why it matters:

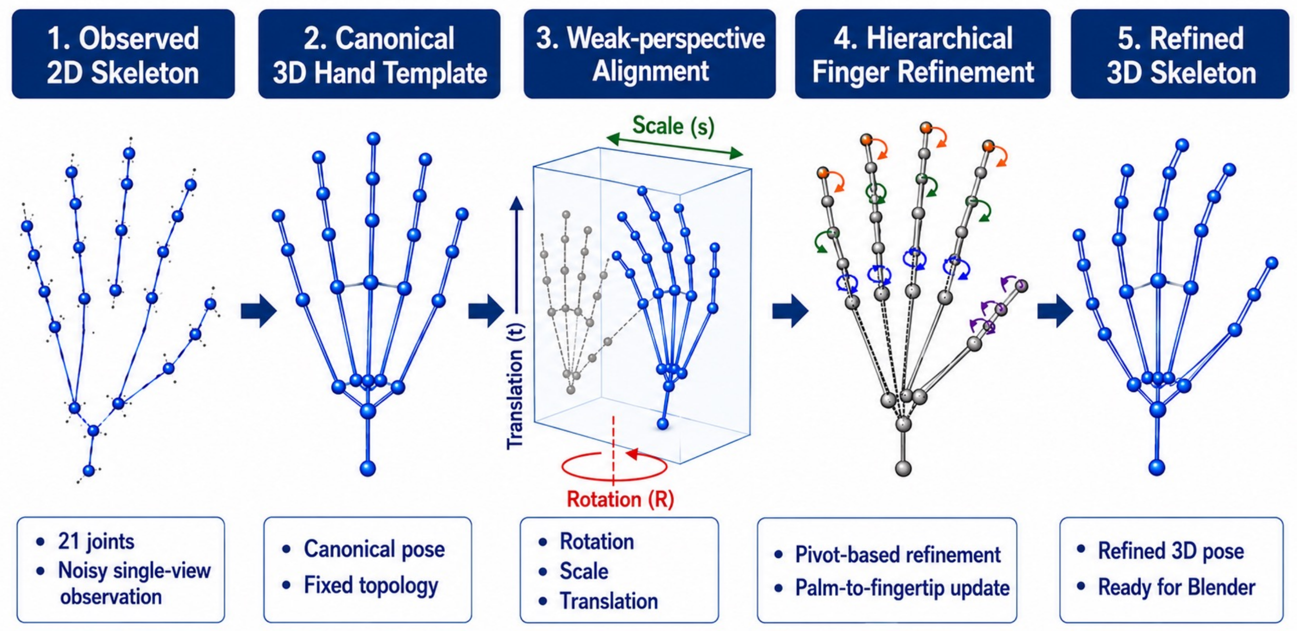
Hand-in-the-wild gestures vary mainly by viewpoint, not only by color, crop, or image rotation.



→ Synthetic multi-view increases viewpoint diversity, not only image appearance diversity

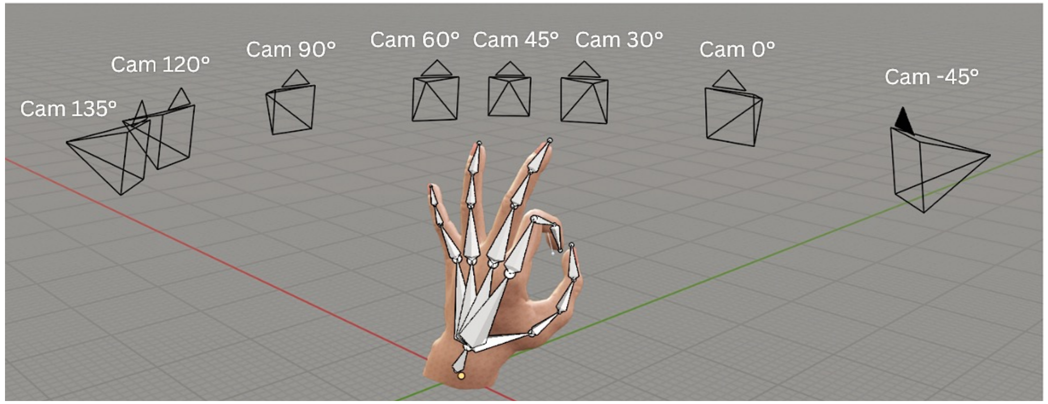
3.3 Reconstruction and Synthetic Multi-view

Geometry reconstruction

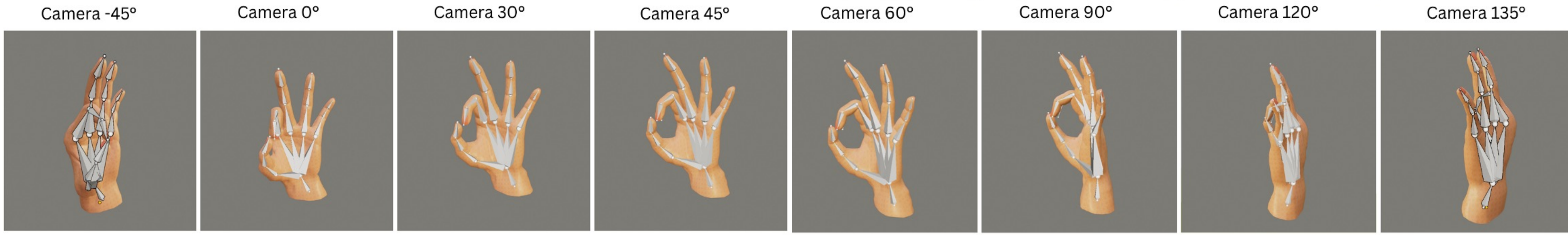


Blender multi-view generation

- Retarget refined skeleton to Blender hand rig
- Render the same pose from 8 virtual cameras
- Extract camera-space skeletons



Rendered Views

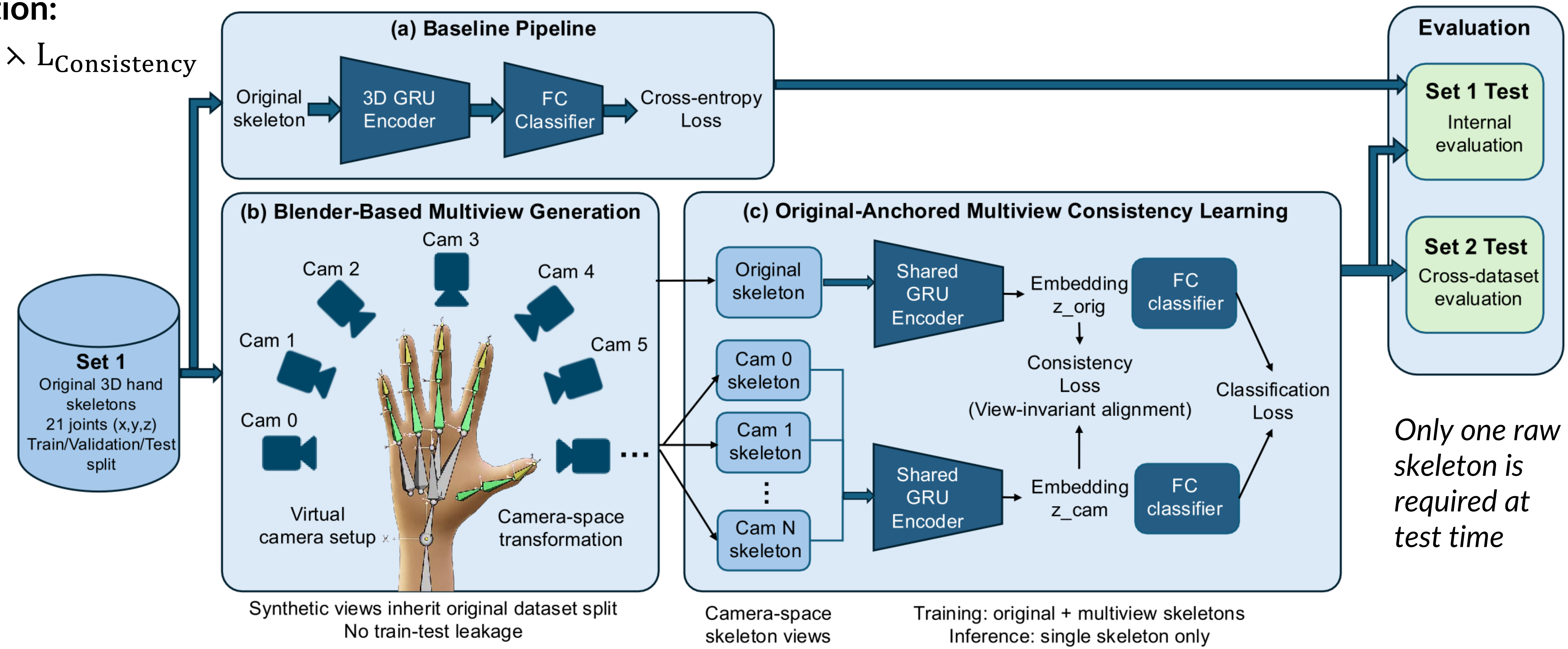


3.4 Multi-view representation learning

Training Strategy: Each original skeleton is paired with multiple synthetic viewpoints.

Loss Function:

$$L = L_{CE} + \lambda L_{Consistency}$$



Only one raw skeleton is required at test time

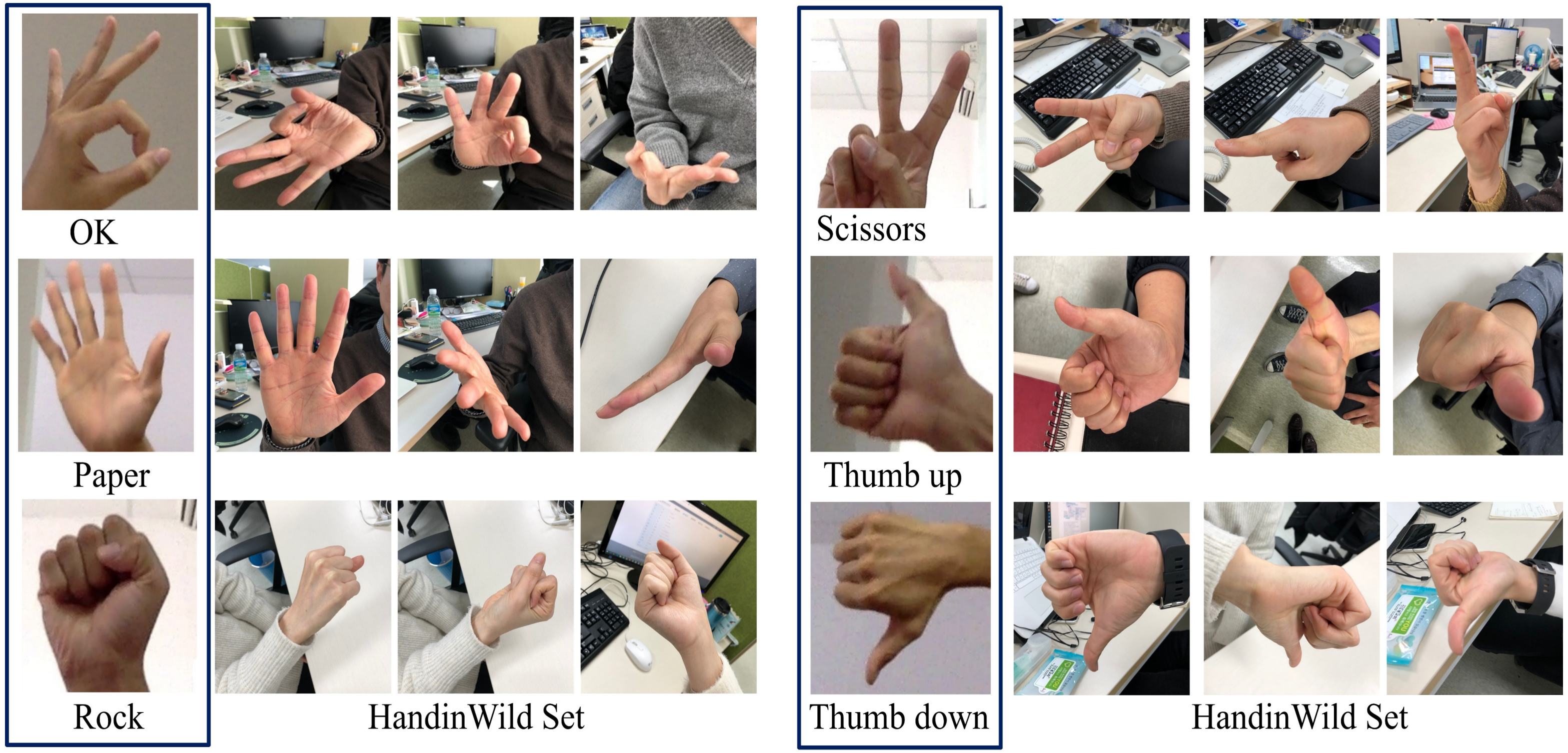
4. Experimental Evaluation

4.1 Experimental setup: Cross-dataset Evaluation

Task: 6-class static hand gesture recognition OK, Paper, Rock, Scissors, The-finger, Thumb

CanonicalSet
- Source dataset
- Training / validation
- Synthetic generation

HandinWildSet
- External test only
- Natural viewpoints
- Uncontrolled background, lighting, and camera conditions



Goal: Evaluate cross-dataset robustness under natural hand-in-the-wild conditions

4. Experimental Evaluation

4.1 Experimental setup

Dataset Statistics

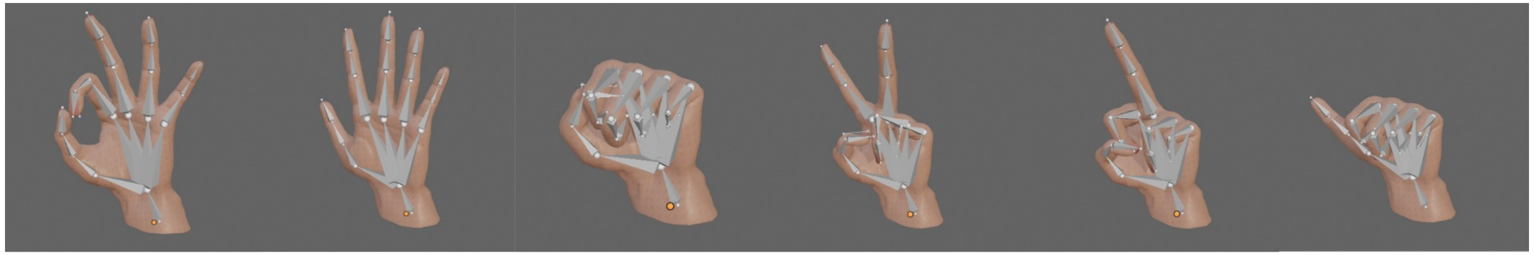
Dataset / Split	Number of Samples
CanonicalSet Train	4,582
CanonicalSet Validation	982
CanonicalSet Test	983
CanonicalSet Total	6,547
HandinWildSet External Test	605
Synthetic Camera-space Views	52,376

Training Configuration

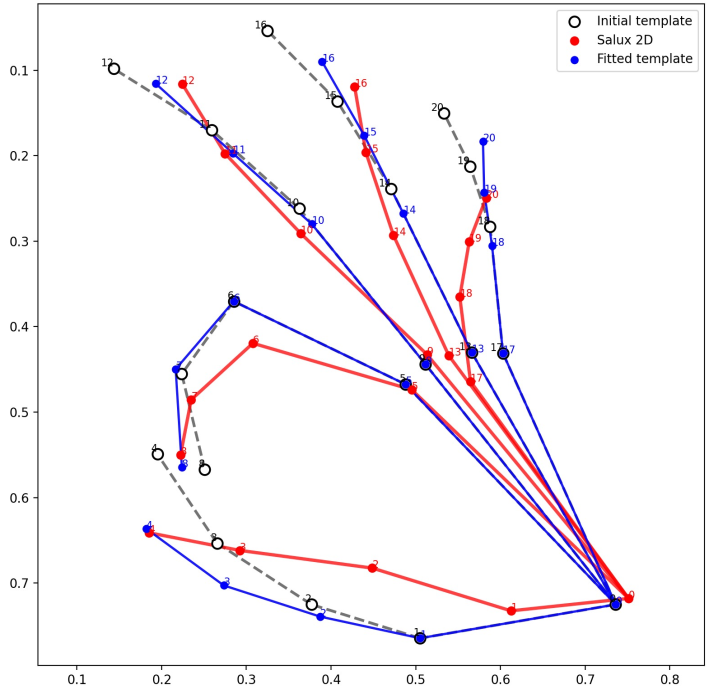
Parameter	Value
Model	GRU
Input shape	21×3
Hidden size	128
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	64
Max epochs	150
Synthetic views	8
Main λ	0.3

4. Experimental Evaluation

4.2 Qualitative results



Ok Paper Rock Scissors The-finger Thumb



Synthetic views preserve hand structure while changing camera viewpoint.

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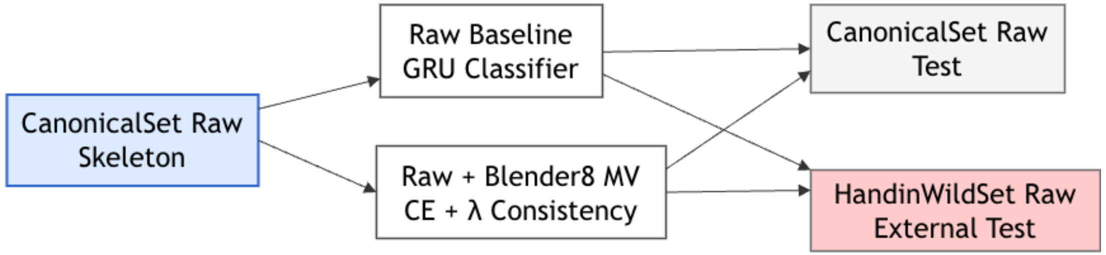
4. Experimental Evaluation

4.3 Quantitative Results

Protocol	Training Input	HandinWildSet Accuracy	Purpose
Raw baseline	Raw skeletons	80.86 ± 2.78%	Deployable baseline
Raw + Blender-8 MV CE-only	Raw + 8 views	84.86 ± 1.35%	Effect of synthetic multi-view
Raw + Blender-8 MV + Consistency	Raw + 8 views	**86.88 ± 1.40%**	Final deployable model
Geometry-refined + Blender-8 MV	Geometry-refined skeleton + 8 views	89.95 ± 3.79%	Refinement upper-bound



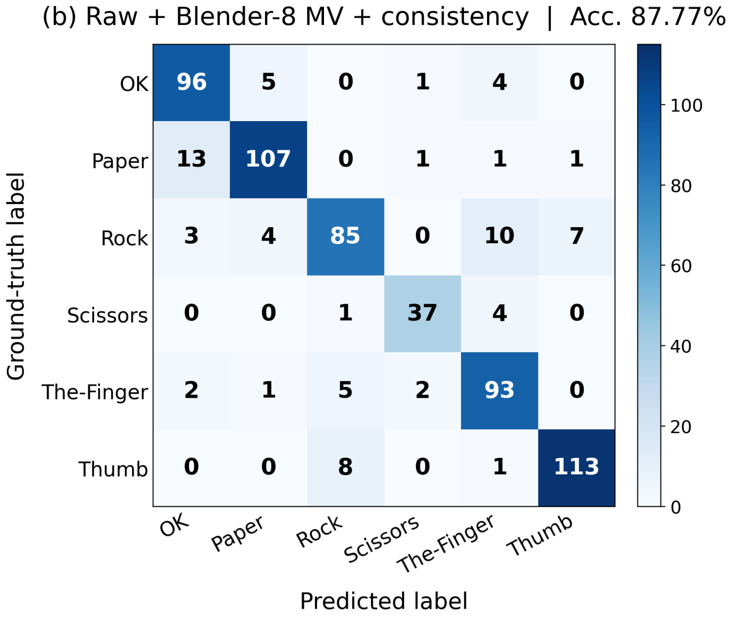
$$\textbf{Accuracy} = \frac{\text{Correct predictions}}{\text{Total test samples}}$$

Results are reported as mean ± std over **5 independent training seeds.**

4. Experimental Evaluation

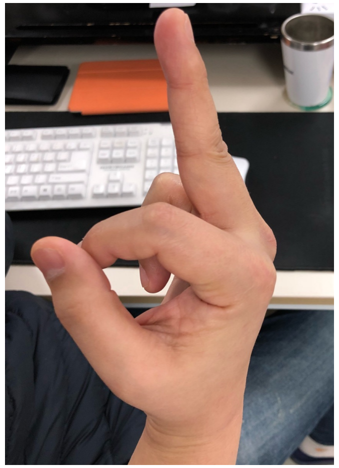
4.4 Failure cases and Limitations

- Similar finger configurations
- Finger self-occlusion
- Noisy landmark estimation
- Non-canonical hand viewpoints



True: Rock
Pred: The-finger
Reason: similar folded finger

True: Ok
Pred: Paper
Reason: open-palm ambiguity

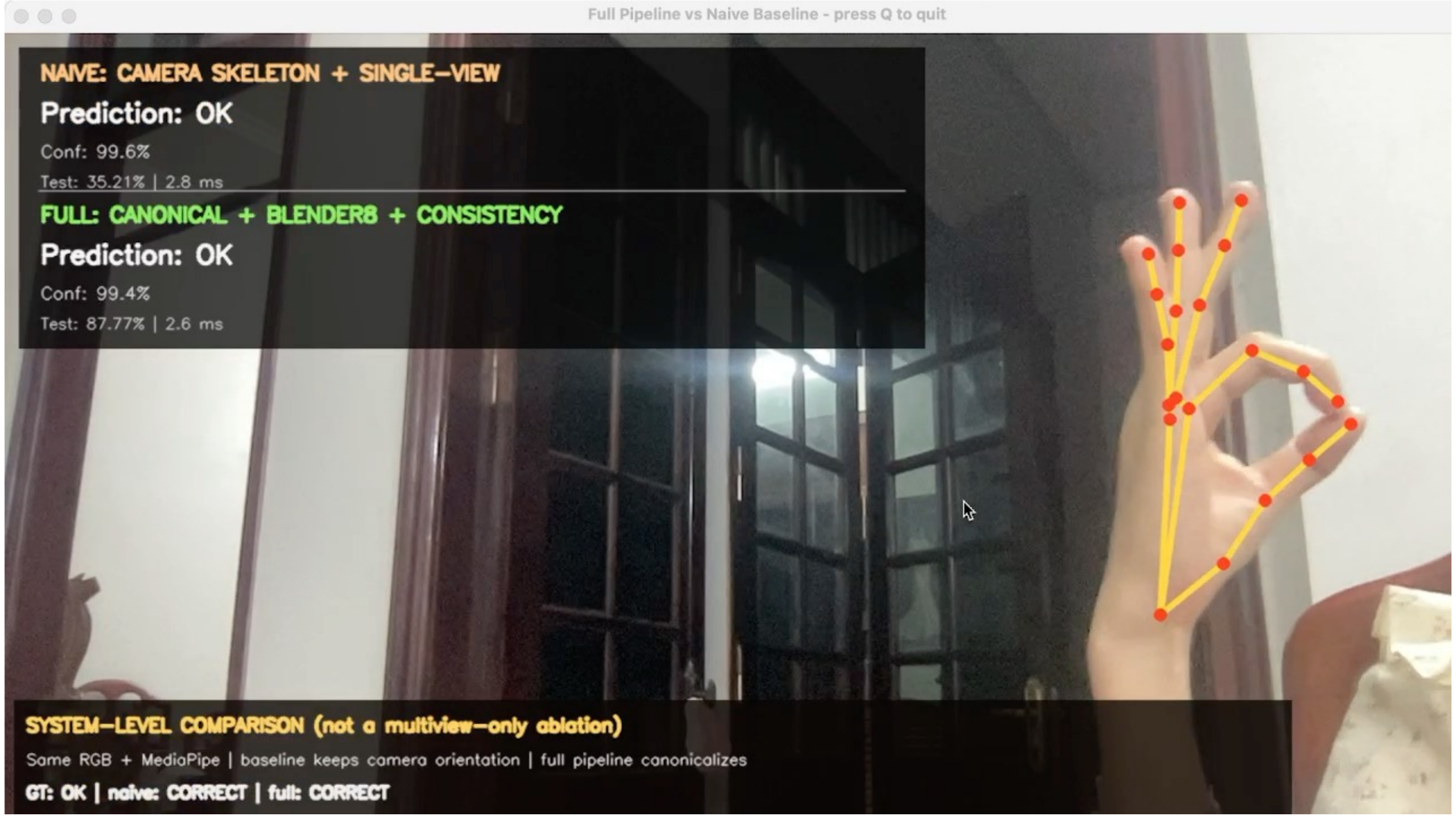


True: The-finger
Pred: Scissors
Reason: partial finger visibility

True: Thumb
Pred: Rock
Reason: thumb visibility

4.5 Short Demo

The demo illustrates practical behavior under natural hand poses.



5. Conclusions and Future work

Main conclusions

- Developed a **geometry-driven synthetic multi-view** pipeline for **hand-in-the-wild** gesture recognition.
- Improved raw-skeleton recognition on HandinWildSet external test set:

  80.86 ± 2.78% → 86.88 ± 1.40%
- Preserved practical **single-skeleton inference**: synthetic multi-view observations are used only during training.

Future work

- Robust automatic geometry refinement / template selection
- RGB-based recognition with better synthetic-to-real adaptation

6. References

Multiview Consistency Learning with Synthetic Camera Views for Hand-in-the-Wild Gesture Recognition

Mai-Huong Nguyen[1], Hai Vu[1], Huong-Giang Doan[2], Thi-Thu-Huong Nguyen[3], and Hoang-Duong Le[1]
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***Abstract*—Hand gesture recognition is an important problem in human–computer interaction, virtual reality, augmented reality, robotics, and multimedia systems. Despite strong performance in previous studies, recognition in uncontrolled environments remains challenging due to viewpoint variation, self-occlusion, articulation ambiguity, and cross-dataset domain shift. To address these issues, we propose a 3D skeleton-based synthetic multi-view framework. A Blender-based animation pipeline reconstructs hand poses from 3D skeleton annotations, while multiple virtual cameras generate camera-space skeleton views from different viewpoints. These synthetic views are used only during training, preserving single-view inference. The original skeleton and its generated views are processed by a shared GRU encoder, and an original-anchored consistency loss aligns their representations. Experiments are conducted under a challenging 7-class cross-dataset setting, using *CanonicalSet* for training and *HandinWildSet* for external evaluation. The proposed 8-camera original-anchored consistency framework achieves the highest accuracy of 75.85% on HandinWildSet, outperforming the GRU baseline (62.96%) and rotation-based augmentation (70.22%). The corresponding 8-camera model without consistency achieves 68.44%, confirming the additional benefit of original-anchored consistency. Experimental results demonstrate that combining synthetic camera-space views with consistency learning improves viewpoint robustness and cross-dataset generalization.**

***Index Terms*—hand gesture recognition, 3D skeleton, multiview learning, Blender, GRU**

I. INTRODUCTION

Hand gesture recognition provides a natural interface for human–computer interaction, virtual reality, robotics, and mul-

Fig. 1: Examples of viewpoint-sensitive hand gesture ambiguity in hand-in-the-wild conditions.

based methods remain sensitive to viewpoint changes because the same gesture can produce markedly different coordinate distributions when observed from different camera perspectives. Consequently, models trained on limited viewpoints may learn viewpoint-specific patterns rather than robust viewpoint-invariant gesture representations.

To address this problem, we propose a multi-view hand representation learning framework that combines synthetic camera-space skeleton generation with original-anchored consistency learning. A Blender-based hand animation pipeline first reconstructs hand articulation from 3D skeleton annotations, after which multiple virtual cameras generate camera-space skeleton observations from different viewpoints. These synthetic views are used only during training as viewpoint su-

Accepted Paper

“ Multiview Consistency Learning with Synthetic Camera Views for Hand-in-the-Wild Gesture Recognition "

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Thi Thu Huong Nguyen, Hoang Duong Le

Accepted at MAPR 2026 – International Conference on Multimedia Analysis and Pattern Recognition

Key References

[1] F. Zhang et al., “MediaPipe Hands: On-device Real-time Hand Tracking,” 2020.
[2] Q. De Smedt et al., “Skeleton-Based Dynamic Hand Gesture Recognition,” CVPRW, 2016.
[3] A. Caputo et al., “SHREC 2021: Skeleton-based Hand Gesture Recognition in the Wild,” 2021.
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